

Disproof of conjecture 00000008419

earthking11

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Verdict: FALSE

We disprove conjecture 00000008419 as stated. The refutation is unconditional and finite: the binary $[5, 2, 3]$ code has dual distance $d^\perp = 2$, and at $t = 1$ its two minimum-weight codewords have supports $\{2, 3, 5\}$ and $\{1, 4, 5\}$, which are not a 1-design. Since the conjecture’s hypothesis $d^\perp \geq t + 1$ holds ($2 \geq 2$) while its conclusion fails, the first clause of the conjecture is false. The same code also falsifies the later clause “the minimal failure of the converse occurs at $d^\perp = 5$ ”: the failure occurs already at $d^\perp = 2$.

1 The conjecture and its exact wording

The statement as filed in `conjectures/00000008419.md` is a compound, partly garbled conjecture. The English text reads:

Definition: The correspondence between covering radius and design strength of codes: the supports of dual codewords of weight w form t -designs. **Conjecture:** For a code with dual distance d-perp at least $t + 1$, all minimal codewords form t -(n, w, λ) designs; the converse of the Assmus–Mattson theorem holds at weight 4, and the minimal failure of the converse occurs at d-perp = 5. (converse of the Assmus–Mattson theorem)

The Chinese text is equivalent; the bilingual original is reproduced verbatim in `README.md`. (It is not typeset here, to avoid shipping a PDF with missing CJK glyphs: this document deliberately uses no CJK package.)

The statement contains three separable assertions:

- (C1) (*main clause*) If a binary linear code has dual distance $d^\perp \geq t + 1$, then all minimum-weight codewords of the code form t -(n, w, λ) designs.
- (C2) The converse of the Assmus–Mattson theorem holds at weight $w = 4$.
- (C3) The minimal failure of the converse occurs at $d^\perp = 5$.

Clause (C1) is false, and it is false already for $t = 1$ and $d^\perp = 2$. Since (C3) asserts that no failure occurs below $d^\perp = 5$, our witness at $d^\perp = 2$ also falsifies (C3). Because the conjecture is the conjunction of its clauses, exhibiting one false clause refutes it as filed.

Remark 1 (on “minimal codewords”). Two readings of “minimal codewords” are natural: the *minimum-weight* codewords, and the codewords whose support is minimal under inclusion. Both readings are falsified by the witness below; see Section 4. We take the minimum-weight reading in

the main argument, since only then do all blocks have a common size w and the phrase “ t -(n, w, λ) design” is well defined.

2 Background: the Assmus–Mattson theorem

The Assmus–Mattson theorem is *not* contradicted by the witness; it is the conjectured converse that fails. For reference, the theorem (binary case) states: let C be an $[n, k, d]$ code over $\mathbb{F}_2$ with dual distance $d^\perp$, and let t be an integer with $1 \leq t \leq d^\perp - 1$ such that the number of weights of $C^\perp$ in $\{1, \dots, n - t\}$ is at most $d - t$. Then, for every w , the supports of the codewords of C of weight w form a t -design.

The extra hypothesis — few dual weights in a short interval — cannot be dropped. For our witness C we have $d = 3$, $d^\perp = 2$, and the nonzero weights of $C^\perp$ are $\{2, 3, 4\}$. For $t = 1$ the AM condition would require $\#\{w \in \{1, \dots, 4\} : w \text{ a weight of } C^\perp\} \leq d - t$, i.e. $3 \leq 2$, which fails. So the AM theorem simply does not apply to this code, exactly because its weight-count condition is violated. Clause (C1) proposes to replace that condition by the much weaker hypothesis $d^\perp \geq t + 1$, and it is this strengthening that is false.

3 The main witness: the binary $[5, 2, 3]$ code

Convention. Coordinates are $1, \dots, 5$, and a word is written as a string $b_1b_2b_3b_4b_5$ whose i -th character is the entry at coordinate i .

Definition 2 (The code C). Let

$$C = \text{span}_{\mathbb{F}_2}\{01101, 10011\} = \{00000, 01101, 10011, 11110\}.$$

Its two generators have weight 3 and their sum 11110 has weight 4.

Theorem 3. *For the code C above:*

- (i) *the minimum distance is $d = 3$;*
- (ii) *the dual distance is $d^\perp = 2$ (the word 01100, of weight 2, lies in $C^\perp$, and $C^\perp$ has no word of weight 1);*
- (iii) *the minimum-weight codewords are exactly 01101 (support $\{2, 3, 5\}$) and 10011 (support $\{1, 4, 5\}$).*

Taking $t = 1$, the conjecture’s hypothesis $d^\perp \geq t + 1$ holds, since $2 \geq 2$. Nevertheless the two supports are not a 1-(5, 3, λ) design: coordinate 5 belongs to both blocks while each of the coordinates 1, 2, 3, 4 belongs to exactly one.

Proof. Direct computation. The nonzero codewords are 01101, 10011 and 11110, of weights 3, 3 and 4; hence $d = 3$. The generators and their sum give the orthogonality conditions for $C^\perp$:

$$y \in C^\perp \iff \begin{cases} y_2 + y_3 + y_5 = 0, \\ y_1 + y_4 + y_5 = 0. \end{cases}$$

The word 01100 satisfies both (it meets the supports of 01101, 10011 and 11110 in 2, 0 and 2 positions respectively), so $d^\perp \leq 2$. No unit vector e_i is orthogonal to all three nonzero codewords

(each coordinate is non-zero on at least one of them), so $C^\perp$ has no word of weight 1; hence $d^\perp = 2$. The weight-3 codewords are exactly the two generators, so their supports are $\{2, 3, 5\}$ and $\{1, 4, 5\}$.

For the design claim observe that the multiset of point multiplicities is

$$(\mu_1, \dots, \mu_5) = (1, 1, 1, 1, 2),$$

so the multiplicity of coordinate 5 is 2 while that of coordinates 1–4 is 1; a 1-design requires a constant multiplicity λ . (In fact λ would have to equal $bw/n = 2 \cdot 3/5 = 6/5 \notin \mathbb{Z}$, which is already impossible.) $\square$

Corollary 4. *Conjecture 00000008419 is false: at $(t, d^\perp) = (1, 2)$ clause (C1) fails, and the failure occurs at $d^\perp = 2 < 5$, so clause (C3) fails too.*

Remark 5 (relation to the proposed counterexample). The refutation proposed in the task statement lists the same code as $\{00000, 01111, 10110, 11001\}$ with minimum-weight supports $\{2, 3, 5\}$ and $\{1, 4, 5\}$. That listing reads each binary string with the *last* character as coordinate 1; reversing every word ($i \mapsto 6 - i$) turns it into the present listing $\{00000, 01101, 10011, 11110\}$ with the same supports and the same multiplicities. The two descriptions are the same code up to coordinate reversal, so the proposal’s conclusion is correct. We use the coordinate-first convention throughout for consistency with the tables and the Lean artifact.

4 Robustness: further witnesses and alternative readings

The failure does not depend on the minimum weight 3, on $t = 1$, or on $d^\perp = 2$. Table 1 records four witnesses and one control; all quantities are computed by `reproduce.py` from the generator matrices over $\mathbb{F}_2$.

generator rows		d	$d^\perp$	t	design property of the minimum-weight codewords
W1	01101, 10011	3	2	1	not a 1-design; multiplicities (1, 1, 1, 1, 2)
W2	1110001, 1001110	4	2	1	not a 1-design; multiplicities (2, 1, 1, 1, 1, 1)
W3	Hamming [7, 4, 3]	3	4	3	1- and 2-designs, but not a 3-design
W4	110001, 101010, 011100	3	3	2	1-design, but not a 2-design
W5	extended Hamming [8, 4, 4]	4	4	3	<i>is</i> a 3-design (control)

Table 1: Four witnesses (W1–W4) to the failure of clause (C1), and one control (W5) on which the design property does hold. Every t in the table satisfies the conjecture’s hypothesis $d^\perp \geq t + 1$.

Reading (a): $d^\perp \geq t + 1$ **for the t at which the design is claimed.** This is the reading used above; W1 (with $t = 1$) refutes it.

Reading (b): “minimal codewords” = minimal-support codewords. For W1 all three nonzero codewords have supports minimal under inclusion: the two weight-3 supports $\{2, 3, 5\}$, $\{1, 4, 5\}$ and the weight-4 support $\{1, 2, 3, 4\}$; no one contains another. Grouping by weight, the weight-3 minimal codewords are again the two generators and are still not a 1-design. For W2 no support contains another either, so all three nonzero words (weights 4, 4, 6) are minimal-support;

grouped by weight, the weight-4 minimal-support codewords are 1001110 and 1110001, with supports $\{1, 4, 5, 6\}$ and $\{1, 2, 3, 7\}$, which again fail the 1-design property. So this reading is false as well.

Reading (c): restrict to weight 4. W2 has minimum distance $d = 4$, $d^\perp = 2$, and its two weight-4 codewords have supports $\{1, 4, 5, 6\}$ and $\{1, 2, 3, 7\}$: coordinate 1 has multiplicity 2 and the others multiplicity 1, so they are not a 1-design. Hence even a reading that confines clause (C1) to weight 4 is false. The control W5 shows that this is not a defect of the design test: the extended Hamming $[8, 4, 4]$ code has $d = d^\perp = 4$ and its 14 weight-4 codewords *do* form a 3-(8, 4, 1) design.

Reading (d): $t \geq 2$. W4 has $d^\perp = 3$, so $t = 2$ is permitted, and its four minimum-weight codewords form a 1-design but not a 2-design. W3 has $d^\perp = 4$ and fails at $t = 3$.

No reading of “ $d^\perp$ at least $t + 1$ ” that keeps the phrase “all minimal codewords form t -(n, w, λ) designs” leaves the statement true: each of the readings above has an explicit counterexample.

5 Reproducibility

`reproduce.py` (Python 3 standard library only) builds each code from its generator matrix over $\mathbb{F}_2$ using integer bitsets, computes the minimum and dual distances, enumerates the minimum-weight codewords and their supports, and tests the t -design property (all t -subsets contained in the same number of blocks). It prints PASS and exits 0 if and only if every check holds; in particular it verifies

```
W1: d = 3, d^perp = 2, supports {2,3,5}, {1,4,5},
    multiplicities (1,1,1,1,2), 1-design = False;
W1 violates the AM weight condition (3 > 2 for t = 1),
W2: minimum weight 4 and d^perp = 2, 1-design = False,
W3: d^perp = 4, 3-design = False,
W4: d^perp = 3, 2-design = False,
W5: control, 3-design = True.
```

6 Formalisation

The witness W1 is formalised in the accompanying Lean 4 project under `lean4/`. It uses core Lean only (`import Std`, no `Mathlib`) and contains no `sorry`. Words are `List Bool` of length 5; XOR is a hand-written Boolean XOR so that the kernel reducer can evaluate it (rather than the opaque `Nat.xor`); the design predicate counts, for every t -subset of the five coordinates, how many blocks contain it, and tests that all counts agree. The project proves, among others,

```
theorem minWeight_eq_three : minWeight = 3
theorem dualDistance_eq_two : dualDistance = 2
theorem support_gA : (List.finRange 5).filter (support5 gA) = [1,2,4]
theorem support_gB : (List.finRange 5).filter (support5 gB) = [0,3,4]
theorem multiplicity_five : countBlocks blocks [4] = 2
theorem multiplicity_one : countBlocks blocks [0] = 1
theorem isTDesign_one_false : isTDesign 1 blocks = false
theorem conjecture_00000008419_false :
```

```
dualDistance = 2 ^ minWeight = 3 ^ isTDesign 1 blocks = false
^ countBlocks blocks [4] = 2 ^ countBlocks blocks [0] = 1
```

Here `support5 gA = [1,2,4]` encodes the support $\{2,3,5\}$ and `support5 gB = [0,3,4]` encodes $\{1,4,5\}$ (the lists contain zero-based coordinate indices). `Check.lean` prints `#print axioms` for every theorem; the audit reports at most `propext` and never `sorryAx`.